

Individual Assignment 01

Observability in the Wild

Adding OpenTelemetry to nopCommerce

Francisco José Gomes Pinto-113763

Table of contents

01

The Problem

02

Architecture

03

Metrics

04

Privacy
Strategy

05

Trade Offs

06

Demo

The Problem

What Standard Monitoring Misses

Traditional monitoring lacks business context:

HTTP 200 hides failures — empty results, failed discounts, and missing data all look healthy.

Revenue-impacting user behaviors go untracked:

Out-of-stock views, zero-result searches, and silent discount failures are invisible revenue losses.

What we need

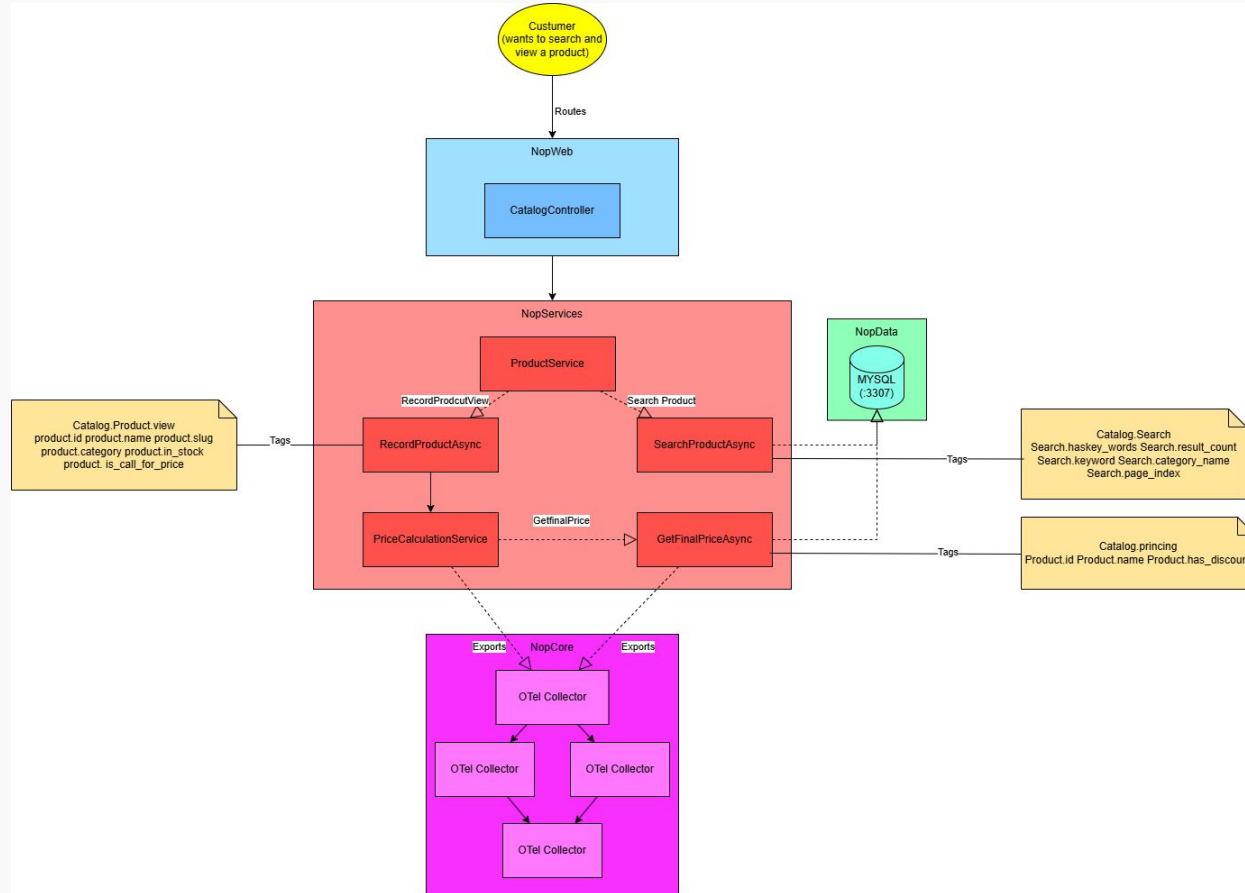
Deep, business-aware observability:

Custom spans at the service layer, enriched with stock status, category, and discount context.

Powerful tracing without code intrusion:

Traces searchable by product name, slug, and category in Jaeger with zero changes to business logic.

Architecture



Metrics

`nop.catalog.searches [found_results]`: Zero-result rate spike = missing catalogue content

`nop.catalog.product_views [product_category]`: Count the views per category of product

`nop.catalog.product_views [in_stock]`: Out-of-stock views return HTTP 200 which leads to lost revenue

`nop.catalog.product_views [is_call_for_price]`: Hidden prices can lead to less traffic

`nop.catalog.pricing_requests [has_discount]`: Enables better pricing strategy and customer insights

Privacy Strategy

Code layer

Only include operational attributes, where each tag represents a deliberate and meaningful choice.

SDK layer

Strips customer.email, billing.*, cookies, db.statement before export.

Collector layer

The second filter in the OTel Collector applies to all telemetry sources. It captures data from auto-instrumentation components outside our direct control.

Trade Offs

Deliberately excluded



EventPublisher consumers

Using `EngineContext.Current` as a static service locator breaks the DI decorator pattern and silently swallows exceptions, making failures hard to detect and debug.



Customer identity in any span

Data collection is limited to operational needs, some information is explicitly excluded to minimize unnecessary data processing and reduce privacy risk.

What I would add next



Product engagement time

Server spans terminate when the HTTP response is sent, not when the user finishes viewing the content, capturing actual user engagement requires a JavaScript beacon endpoint.



More custom metrics

Add `nop.catalog.search.page_depth` for diagnosing how relevant the primary information is

Demo



Thanks!

Francisco José Gomes Pinto-113763
24/03/2026